

Critique and Improvement Plan for SIH26057: An End-to-End Side-Scan Sonar Marine-Debris Detection Pipeline

TL;DR

The single biggest risk in this build plan is real annotated data scarcity for the specific target classes (ghost nets, entangled gear). No public side-scan sonar (SSS) dataset is labeled for ghost nets; the largest usable SSS resources (SeabedObjects-KLSG = 1,190 images, SCTD ~357 public, AI4Shipwrecks 286, Marine-PULSE ~627) cover wrecks, planes, mines, pipelines/cables and terrain — not entangled fishing gear. The team must reframe "ghost-net detection" as a small-data / transfer-learning / synthetic-augmentation / anomaly-detection problem or it will overfit and fail on judges' held-out data.

The proposed stack (YOLO11-n/s + EfficientNet-B0 verifier, CLAHE/speckle denoising, geotagging, edge deployment) is reasonable and competitive — it mirrors 2025 SSS literature — but it should be upgraded with sonar-specific preprocessing (slant-range correction, TVG/beam-pattern normalization, wavelet/Lee-Frost speckle filtering), shadow-aware features, GAN/diffusion synthetic data, and honest geolocation error bars. A two-stage detector+verifier is a smart way to cut the false-alarm rate that plagues SSS.

To win, the team must nail what SIH judges actually score — novelty, feasibility, problem understanding tied to the MoES use case, and a working demo — and be honest about limitations (positional accuracy of several meters; domain shift across sonar frequencies). A polished pipeline that ingests real XTF/survey data, geotags contacts, and exports JSON/CSV with calibrated confidence will beat a higher-mAP model with no deployable workflow.

Key Findings

Problem statement. SIH26057 is a Software problem statement titled "AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery," sponsored by the **Ministry of Earth Sciences (MoES)**, theme "Smart Automation," prize ₹1,00,000. The specific MoES sub-institute is not confirmed publicly, but the subject matter aligns most closely with NIOT (Chennai). MoES has several sibling ocean-tech PS in 2026 (SIH26058 software-defined sonar payload for AUVs; SIH26064 seafloor metal detection). SIH 2026 (9th edition) launched August 21, 2026 with 226 problem statements (172 software + 54 hardware) across 30 organizations; "Smart Automation" is one of the largest themes (31 statements).

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Data reality. There is enough public SSS data to build a credible wreck/plane/pipeline/mine detector, but NOT a robust, dedicated ghost-net detector. Ghost-net-specific work (GhostNetZero, GhostVision, EvoLogics, sonoware GhostNetBusters) uses private datasets.

Methods. YOLOv8/YOLO11 variants dominate 2025 SSS detection literature; transformer/RT-DETR, SAM/SAM2 fine-tuning, and diffusion-based augmentation are the current research frontier.

Deployment & geotagging. Edge deployment on Jetson Orin with TensorRT INT8/FP16 is well-established; SSS geotagging via XTF metadata + slant-range + layback is standard but accuracy-limited to the meter level.

Details

1. Real, publicly available side-scan sonar datasets

SeabedObjects-KLSG (Hohai University, Huo, Wu & Li, IEEE Access 2020, vol. 8, pp. 47407–47418). The most cited real SSS dataset. Full set: **385 shipwrecks, 36 drowning victims, 62 airplanes, 129 mines, 578 seafloor images = 1,190 total**, imaged with suppliers L-3 Klein, EdgeTech, Lcocean, Hydro-tech Marine, and Trittech; resolutions range from 72×52 to 2374×3092 px. A subset of 1,171 images (excluding mines) is public for academic use. Originally **classification-level labels only (no bounding boxes/masks)** — later works manually re-annotated it for detection (e.g., one JMSE 2025 study used 447 public images: 395 ship + 62 aircraft, hand-drawing bounding boxes). Highly class-imbalanced; no GPS metadata. Access via request to authors / mirrored repos. [arXiv](#)

SCTD 1.0 (Sonar Common Target Detection, multi-university). 596 targets across 3 classes; ~357 images publicly accessible (57 planes, 266 shipwrecks, 34 drowning victims). **Pascal VOC XML bounding-box annotations** (with a voc2coco converter). Includes SSS, FLS and SAS imagery. Available on GitHub (MingqiangNing/SCTD). Small, dimensionally heterogeneous.

AI4Shipwrecks (University of Michigan, Sethuraman/Sheppard et al., IJRR 2025; arXiv 2401.14546). **28 distinct shipwrecks totaling 286 high-resolution SSS images with pixel-wise binary segmentation masks.** Collected over five weeks in 2022–2023 at NOAA Thunder Bay National Marine Sanctuary, Lake Huron, via an Iver3 AUV with an EdgeTech 2205 dual-frequency ultra-high-resolution SSS. Expert marine-archaeologist labels. Open-source (Deep Blue Data, DOI 10.7302/dmf4-x492; project page umfieldrobotics.github.io/ai4shipwrecks), pre-split 50/50 train/test. This is the best public dataset for segmentation + anomaly detection and the closest analog to the SIH task. It is hard — a 2025 model (SW-Net) reaches only IoU 39.43% / F1 56.56%. [Semantic Scholar](#)

Marine-PULSE (Du et al. 2023). ~627 SSS images of marine engineering structures: 323 pipelines-or-cables (POC), 134 underwater residual mounds (URM), 82 engineering platforms (EP), 88–180 seabed surface (SS). Directly relevant to the "pipes/cylinders" target class. Classification labels; no GPS. On Zenodo (record 7922705). Collected with EdgeTech4200FS, Benthos SIS-1624, Klein-2000/3000. A 2025 study reports YOLO11n-seg with A2C2f + DSConv modules doing real-time pipeline/cable segmentation on this data.

SWDD (Sonar Wall Detection Dataset, OceanScan/REMARO). SSS from a Klein 3500 on a light AUV in Porto de Leixões harbor. Original ~216 raw images augmented; the sonar-dataset survey lists ~7,904. The extended SWDD-Validation set (SWDD-Clean/Surface/Noisy) on Zenodo (record 10528135) carries bounding boxes plus sonar frequency, range and resolution metadata — useful for robustness/domain-shift testing. Class: walls (not debris) but valuable for methodology (e.g., ROSAR adversarial re-training).

SubPipe (SSS, ~10,030 images). Pipeline detection with pose data for SLAM; bounding boxes + metadata.

SSS Mine Detection (~1,170 images, Teledyne Gavia AUV). Mine-like objects; detection labels; on figshare.

Forward-looking sonar (distinct from SSS). The Marine Debris FLS dataset (Singh & Valdenegro-Toro, ICCVW 2021; arXiv 2108.06800) is the one dataset literally labeled "marine debris" — but it is **forward-looking sonar**. It consists of **1,868 FLS images captured with an ARIS Explorer 3000 sensor, divided into 11 classes plus a background class**; because of its ~15 Hz data rate it acts as an "acoustic camera" with per-pixel resolution up to 2.3 mm. FLS has shorter range and different sensor geometry than SSS, so a model trained on it will NOT transfer cleanly to towed/AUV SSS survey imagery — use it for concept/augmentation only and be explicit about the FLS-vs-SSS domain gap. UATD (Pengcheng Lab, ~9,200 FLS images with bounding boxes) and the FLS UXO dataset are other FLS resources. [GitHub](#)

Aggregators to cite/use: the GitHub lists "Awesome-Sonar-Image-Resources" (Jorwnpay) and "OpenSonarDatasets" (remaro-network), and the 2025 survey "Sonar Image Datasets: A Comprehensive Survey" (arXiv 2510.03353), which tabulates modality, size, labels and GPS availability for ~25 datasets.

JAMSTEC Deep-sea Debris Database and SeaClear (8,610 images, 40 categories) are optical/ROV camera debris datasets — not sonar — useful only for context.

Verdict on data scarcity: This is the central risk. There is NO public SSS dataset labeled for ghost nets or entangled fishing gear, and most SSS datasets are classification-only, small (hundreds of images), single-object-per-image, class-imbalanced, and lack GPS metadata. The team should (a) assemble a composite training set (KLSG + SCTD + AI4Shipwrecks + Marine-PULSE, re-annotated to a common format for the SIH target classes); (b) treat ghost-net/entangled-gear as an anomaly/novelty class rather than a fully supervised class; (c) heavily use synthetic data; and (d) frame natural features (rocks, ripples, ridges) as an explicit "hard-negative" background class to cut false alarms.

2. State of the art (2025–2026) and critique of the proposed model

Is YOLO11-n/s + EfficientNet-B0 verifier reasonable? Yes — and competitive. YOLOv8/YOLO11 variants are the workhorses of 2025 SSS detection papers: SS-YOLO (lightweight, edge-focused, JMSE 2025), improved YOLOv11 for small SSS objects (JMSE 2025, 13:162), an enhanced YOLOv11 + attention + BiFPN (ABFP-YOLO, JMSE 2025 13:926), ESL-YOLO (edge-aware, JMSE 2025), BES-YOLO (EMA attention + BiFPN + wavelet preprocessing, Sensors 2024), CSC-YOLO (YOLOv8n for wrecks), and CKAN-YOLOv8 (multi-task detection+segmentation). Note that one 2025 study (SOCA-YOLO) reports YOLO11 being beaten by a YOLOv9-based model with super-resolution restoration by +2.3% mAP50 — so YOLO11 is a strong but not automatically top baseline. [doaj](#)

Recommended upgrades to the model plan:

Add a **BiFPN or attention module (CBAM/EMA)** — repeatedly shown to help multi-scale SSS targets. Consider **YOLO11-seg / instance segmentation** rather than pure bounding boxes: ghost nets and debris fields are irregular, and AI4Shipwrecks proves segmentation is the natural task.

The **EfficientNet-B0 second-stage verifier is a good, cheap idea** to reduce false positives (the dominant SSS failure mode) — essentially the classic "detect then classify" that KLSG transfer-learning papers use. A

complement: a shadow-shape verifier, since acoustic shadow is often more discriminative than the highlight. Frontier options to mention for novelty (not necessarily to build): **RT-DETR / transformer detectors** (2026 Frontiers "Lightweight Edge-Frequency Driven Real-Time Detection Transformer"); **SAM/SAM2 fine-tuning** (arXiv 2306.14109 "When SAM Meets Sonar Images"; AquaSAM; 2025 IEEE GRSL fine-tuning SAM for FLS with collaborative prompts) for label-efficient segmentation; **self-supervised/domain-adaptive** pretraining (DCBD, Nature Sci. Reports 2024, ~92% AP50 unsupervised shipwreck detection); wavelet backbones (WPG-DetNet / DWT downsampling, Sensors 2026). [arXiv + 2](#)

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Preprocessing that is actually shown to help in SSS literature:

Slant-range correction (slant → ground range) — a geometric prerequisite that makes object sizes/positions physically meaningful.

Radiometric/intensity normalization: TVG residual correction, beam-pattern correction, z-score/ping-energy normalization (Remote Sensing 2017 9:575).

Speckle denoising: Lee, Frost, SRAD spatial filters; wavelet/2D-DWT (used in BES-YOLO); BM3D/NSCT variants; tensor (Tucker) decomposition; Bayesian sparse coding. Trade-off: contrast enhancement amplifies noise, so denoise *before* contrast stretching.

CLAHE / histogram equalization / gamma / RGHS for gray-balance and contrast — CLAHE (clip-limited) is preferred because plain HE increases noise. Multi-stage fusion enhancement (Electronics 2023 12:3553) combines denoising + gray balance.

Caution: aggressive denoising can erase the shadow/edge cues that distinguish man-made objects from seabed. Validate that preprocessing improves downstream mAP, not just visual quality.

3. Known technical challenges & failure modes (with proven fixes)

Acoustic shadow exploitation. Man-made objects cast characteristic shadows; shadow shape is often more diagnostic than the acoustic highlight and encodes object height. Fix: engineer shadow-aware features / include shadow in masks; region-growing on highlight+shadow (PMC8588333). Judges will be impressed by explicit shadow use.

Slant-range distortion & geometric correction. Horizontal distance in raw sonograms corresponds to slant range, not ground range; the flat-seabed assumption fails on slopes. Fix: bottom tracking → slant-range correction → (ideally) seabed-topography-aware rectification. [uspto](#)

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Heave/pitch/roll/yaw motion artifacts (towfish instability). Cause along-track warping. Fix: attitude correction from IMU/nav data; image-based distortion estimation (Cobra et al.); yaw/pitch geocoding (arXiv 2604.19901). Towed fish are less stable than hull-mounted or AUV sonar. [Academia.edu](#)

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Speckle noise. Multiplicative coherent noise; fixes as in §2.

Class imbalance & tiny datasets. KLSG is severely imbalanced (385 wrecks vs 36 victims). Fixes: offline+online augmentation (flip/rotate/translate, Mosaic/MixUp), transfer learning from ImageNet (pretraining raised accuracy ~10% in the Marine-PULSE study), semi-synthetic and GAN/diffusion data, focal loss, oversampling, and treating rare classes as anomalies.

Domain shift across sensors/frequencies (100–600 kHz; EdgeTech vs Klein vs Marine Sonic). A model trained on one sonar generalizes poorly to another. Fixes: domain adaptation (DCBD), adversarial re-training (ROSAR on SWDD), multi-frequency training, strong augmentation, and honest cross-sensor validation. SIH judges may test on unseen sonar data — reserve a cross-sensor test split.

False positives from natural features (rocks, ripples, ridges). This is THE core requirement of SIH26057. Fixes: explicit hard-negative background class, the EfficientNet verifier stage, shadow-geometry rules, and calibrated confidence thresholds (precision–recall tuned for a low false-negative rate).

4. Synthetic data generation / simulation — a credible hackathon strategy

Synthetic data is mainstream SSS practice and a source of novelty points:

Semi-synthetic / style transfer: paste optical silhouettes and style-transfer to sonar appearance (KLSCG authors' original method; MDPI Electronics 2021 style-transfer + pretrained CNN).

GANs: Markov cGAN full-scale continuous sonar (Jegorova 2020); multi-frequency SSS GAN (Jiang 2020); CycleGAN synthetic sonar (Koo 2024); CC-WGAN with CBAM (Appl. Sci. 2024 14:8031); SIGAN super-resolution.

Diffusion models (current best): a DDPM/DDIM SSS augmentation method reached 91.5% detection accuracy on real data (JMSE 2023 11:1103); Synth-SONAR dual-diffusion + GPT prompting (arXiv 2410.08612); FLUX+LoRA few-shot SSS synthesis; a 2025 diffusion augmentation study (Appl. Sci. 16:7557) found mAP50:95 improved from 0.588→0.600 at a 60% synthetic-injection ratio but degraded beyond that — i.e., there is an optimal synthetic:real mix and a domain-shift boundary. DS-SIAUG (self-training + DDPM + YOLO, ~+5% accuracy).

Simulators: S3Simulator (arXiv 2408.12833) uses SAM to segment optical ship/plane silhouettes → SelfCAD 3D models → Gazebo rendering of sonar noise/shadows/terrain — an open-source pipeline (github.com/bashakamal/S3Simulator) adaptable to synthesize pipes/cylinders/nets.

Recommendation: Use diffusion or CycleGAN to synthesize the scarce classes (nets/entangled gear/cylinders), keep synthetic ≤~50–60% of the training mix, and report an ablation showing the accuracy gain — this doubles as a novelty story for judges.

5. How SSS geotagging actually works (and its accuracy limits)

Pipeline to convert a detected pixel to lat/long:

Towfish/sensor position from GPS. For a towed fish, the GPS is on the vessel, so compute fish position from vessel GPS – offsets – **layback** (horizontal distance the fish trails behind the vessel), typically via Pythagoras from cable-out length and fish depth ($\text{layback} \approx \sqrt{(\text{cable}^2 - \text{depth}^2)}$, minus fairlead offsets). AUV/USBL systems avoid layback error by carrying GPS/INS or acoustic positioning.

Heading/attitude (gyro, pitch, roll, yaw) to orient the swath.

Slant range → ground range for the target's cross-track bin: $\text{ground_range} \approx \sqrt{(\text{slant}^2 - \text{altitude}^2)}$ (flat-bottom assumption), using bottom-tracked altitude.

Along-track position from ping number/time and navigation.

Project to a datum/UTM (e.g., WGS84 / NAD83) to output object lat/long.

Formats: the de-facto standard is **XTF (eXtended Triton Format)** — stores sonar samples plus ship and sensor lat/long, layback, cable-out, heave/pitch/roll, altitude, heading (rev. 26 documented by MBARI). Data acquired via ISIS; processed in CARIS SIPS/HIPS, SonarWiz, or open tools (SonarTRX, Hydromagic, the [pyxtf](#) Python library). USGS/NOAA publish georeferenced GeoTIFF SSS mosaics (e.g., USGS OF 2010-1005, Klein 5000 at 455 kHz).

Accuracy limits (be honest in the report): positional accuracy is typically on the order of **several meters**, dominated by layback uncertainty (for towed fish), heading error, GPS error, sound-velocity/altitude error, and flat-seabed error on sloped terrain. Kalman-filtered multi-pass measurement and attitude-correction matrices improve this. Output a **position + estimated error radius**, not a false-precision single lat/long, and geo-reference to a slant-range-corrected mosaic rather than raw waterfall pixels where possible.

6. What differentiates winning SIH projects + SIH 2026 logistics

SIH 2026 (9th edition) launched August 21, 2026 (AICTE Chairman Prof. Yogesh Singh), organized by the Ministry of Education's Innovation Cell and AICTE; 226 problem statements (172 software + 54 hardware) across 30 organizations. Flow: internal college hackathon (September 2026) → national screening → Grand Finale (a 36-hour hackathon, historically in December). The catalogue lists a 20 September 2026 deadline against SIH26057, though a secondary source cites 30 September for national idea submission — verify the exact date on [sih.gov.in](#). [Reskill + 3](#)

Official evaluation criteria (verbatim, stable across editions incl. SIH 2025): "Evaluation criteria will include novelty (i.e. the originality) of the idea, complexity, clarity and details in the prescribed format, feasibility, practicability, sustainability, scale of impact, user experience and potential for future work progression." Scoring is over three rounds, 1-20 per criterion, weighted to a final /100. A 2025 evaluator characterization lists Innovation, Invention, Technical Feasibility, Impact & Benefits, and Architecture. [Scribd](#)

[CodeHunters Academy](#)

What judges reward (and common student weaknesses):

Deep problem understanding tied to the sponsor. Winners research the real MoES need (marine-debris/anomaly detection for ocean monitoring), not a generic CV demo. Weakness: high mAP on a toy dataset with no link to the actual survey workflow.

Feasibility + a working end-to-end demo. A pipeline that ingests real XTF/survey imagery, runs detection, geotags, and exports JSON/CSV + dashboard beats a stronger model with no product. Weakness: over-scoping, buzzword-stuffing, no live demo, or a demo that only works on cherry-picked images.

Honest evaluation. Cross-sensor test split, precision/recall (not just mAP), false-alarm rate, and stated limitations. Weakness: leaderboard-chasing on a leaked/overlapping test set; ignoring domain shift.

Novelty done right: synthetic-data ablation, shadow-aware features, anomaly detection for unseen debris types, edge deployment with measured FPS/watts.

Robust demo prep: offline demo, recorded backup video, backup files — judging happens under time pressure and flaky Wi-Fi.

Edge deployment specifics to cite: YOLOv8/11-nano on **NVIDIA Jetson Orin Nano** runs real-time; **TensorRT INT8/FP16 quantization** gives $\sim 1.8\times$ (FP16) to $\sim 2.4\times$ (INT8) speedups on Jetson AGX Orin, with FP16 nearly lossless and INT8 a small mAP drop; Orin Nano can exceed 25 FPS in 7–15 W modes. Export via ONNX → TensorRT. This makes the "edge deployment" deliverable very achievable and demo-able — report measured FPS, latency (ms), model size (MB), and power (W). [Springer](#)

Recommendations

Stage 1 — De-risk the data problem first (before touching model architecture):

Assemble a composite, re-annotated dataset in one format (recommend YOLO-seg / COCO): AI4Shipwrecks (masks) + KLSG + SCTD + Marine-PULSE (pipes/cylinders), plus a large **hard-negative "natural seabed" class** (rocks/ripples/ridges) drawn from NNSSS/Seafloor Sediments/AI4Shipwrecks backgrounds.

Define target classes realistically: `wreck`, `pipe/cylinder`, `debris/anomaly` (catch-all incl. nets), `natural-seabed` (negative). Treat ghost nets/entangled gear as an **anomaly/novelty** detection sub-task, not a fully supervised class — and say so explicitly.

Generate synthetic net/cylinder samples via CycleGAN or a diffusion model (or adapt S3Simulator); cap synthetic at $\sim 50\text{--}60\%$ and run an ablation.

Stage 2 — Build a sonar-correct pipeline (not a generic image pipeline): 4. Preprocessing order: bottom-track → slant-range correction → radiometric/TVG/beam-pattern normalization → speckle denoise (Lee/Frost/wavelet) → CLAHE. Validate each step improves downstream mAP. 5. Model: YOLO11-s(-seg) + BiFPN/attention as primary; EfficientNet-B0 (or shadow-shape) verifier as a false-positive-suppressing second stage. Keep YOLO11-n as the edge model. 6. Geotagging: parse XTF (pyxtf), implement layback + slant-to-ground + attitude projection to WGS84; output lat/long **with an error radius**.

Stage 3 — Make it demo- and judge-ready: 7. Dashboard + JSON/CSV export per contact: class, calibrated confidence, lat/long $\pm$ error, shadow-derived size estimate, thumbnail. 8. Deploy the quantized model to Jetson Orin (or simulate); report FPS/latency/size/power. 9. Evaluate on a **cross-sensor held-out split**; report precision, recall, false-alarm rate, mAP; show failure cases honestly. 10. Presentation: lead with the MoES problem, show live + backup demo, feature the synthetic-data and shadow-aware novelty story, and a roadmap for real field validation.

Benchmarks / thresholds that should change the plan:

If composite real+synthetic training still yields recall $< \sim 0.7$ on held-out debris/anomaly, **pivot from detection to anomaly/segmentation + human-in-the-loop review** (mirrors GhostNetZero's operational framing of reducing verification dives).

If cross-sensor mAP drops $> \sim 15\text{--}20\%$ vs in-domain, prioritize domain adaptation / stronger augmentation over squeezing in-domain accuracy.

If Jetson inference $< \sim 15$ FPS at target resolution, drop to YOLO11-n + INT8 and/or tile the imagery.

Caveats

The exact official SIH26057 description and expected deliverables were not retrievable from public sources. The metadata (MoES, Software, Smart Automation, ₹1 lakh) is confirmed via the SIH 2026 catalogue, but the full ministry background/"Expected Solution" text and the specific MoES sub-institute are unconfirmed — read the authoritative version on sih.gov.in (the full description field exists on the per-PS detail page).

SIH 2026 dates conflict across secondary sources (20 Sept vs 30 Sept 2026 idea-submission deadline; a December Grand Finale is stated in expected/future tense). Verify on the official portal.

Dataset counts vary between sources because papers use different public subsets and re-annotations (e.g., KLSG "1,190 total" vs "1,171 public" vs "447 detection-labeled"; SCTD "596 targets" vs "357 public images"). Treat these as approximate and cite the specific subset used.

Benchmark numbers (e.g., SW-Net IoU 39.43%, DCBD ~92% AP50, diffusion 91.5%) come from individual papers on their own splits and are not directly comparable to what the SIH team will achieve on a composite dataset.

Several cited methods are research prototypes, and some claims (e.g., GhostNetZero's "94% accuracy," YOLO26 edge benchmarks) are vendor/author-reported and not independently verified.

The FLS "marine debris" dataset (1,868 images, 11 classes + background) is real but is **forward-looking, not side-scan** — do not treat its results as transferable to SSS without domain-gap caveats.